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Fill the corner, don't fight it: how pendentives put a dome on a square

The free guide to the reel on Hagia Sophia: the problem of a round dome over a square room, how pendentives solve it, the story of the 558 collapse and the rebuild, and a paper-and-orange exercise that makes four pendentives.



A dome in under six years

Hagia Sophia in Constantinople (now Istanbul) was begun on 23 February 532 and inaugurated on 27 December 537, five years and ten months later. It was designed by Anthemius of Tralles and Isidore of Miletus.

The result is one of the first buildings to use a fully pendentive dome. After the dome's partial collapse in 558, the rebuilt dome has a diameter of about 32.7 to 33.5 m and an interior height of about 55.6 m.

The problem

A dome is round. A room is square. Where a circle meets the four corners of a square there is a gap: you cannot simply sit a hemisphere on four walls, because it would hang over the corners.

Builders solved this in several ways: cutting corners with small arches (squinches), or filling the gap with a smooth curved surface.

The solution: pendentives

Imagine a sphere that passes through the four corners of the square room. Cut it with four vertical planes along the walls, then cut it horizontally at the top of the arches. What remains at the corners are four curved triangles: pendentives. The ring above them is a circle that fits exactly inside the square.

The dome sits on that ring, and the pendentives carry its weight down to the four corner piers. This is the idea in the reel's diagram.

In Hagia Sophia the pendentives are decorated with seraphim, six-winged figures, which you can see in the corners under the dome.

The collapse and the rebuild

An earthquake in 558 brought down part of the structure; the vault was judged to have been too flat and overloaded. It was rebuilt about 20 Byzantine feet (around 6 m) higher, and the church was re-dedicated on 24 December 562. In 1317 four new buttresses were added on the eastern and northern sides.

Built examples

Hagia Sophia, Istanbul (532-537)	A pendentive dome over a square bay, with half-domes on either side.
Muqarnas domes	Another way to bridge a square to a round shape: stepped tiers of small cells.
Renaissance and modern domes	Many later domes, including the Pantheon-style ones built on drums, use similar transitions.

Try it yourself: make four pendentives

1. Draw a 10 cm square on card: your room. Mark each side's midpoint.
2. Take a round fruit, like an orange: your sphere. Ask it to rest on the four corners of your square. It cannot sit flat, because the circle through the corners is larger than the square.
3. Cut the orange peel into a dome and trim it with four vertical cuts that follow the square's sides: you now have a cut sphere with four straight edges.
4. Observe the four corner pieces: they are curved triangles, the pendentives.
5. Remove the top cap: the ring that remains is a circle that fits inside your square, ready for a dome.

The rule to remember

When two shapes will not meet, fill the gap with a surface that moves smoothly from one to the other.

Default answer	Cut the corners off one of the shapes.
Form-finder answer	Add a transition surface that carries the load across.
Check	Ask: where do two of my shapes meet badly, and what surface could join them?

Go further

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https://en.wikipedia.org/wiki/Hagia_Sophia

Pendentive (Wikipedia)
https://en.wikipedia.org/wiki/Pendentive

Anthemius of Tralles (Wikipedia)
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